

BREATHING & CHEST PROBLEMS

Respiratory Distress | Asthma | Pneumonia | Pneumothorax | Chest Wounds

Clinical Guideline for AI Model Training Data — Version 1

Sources: WHO IMAI | MSF Clinical Guidelines | Hesperian Health Guides | ICRC War Surgery | WHO ARI Guidelines

Respiratory emergencies are among the most rapidly fatal conditions in any setting. Correct patient positioning, airway management, and early recognition of life-threatening chest injury are critical skills in conflict and low-resource environments where mechanical ventilation is unavailable.

1. RECOGNISING RESPIRATORY DISTRESS — UNIVERSAL SIGNS

Sign	Description	Severity Indicator
Respiratory rate	Normal adult: 12–20 breaths/min. Count for full minute.	< 10 or > 30 = danger
Accessory muscle use	Neck muscles, shoulder muscles, or abdominal muscles straining with each breath	Sign of significant distress
Nasal flaring	Nostrils widening with each breath — especially significant in children	Moderate–severe distress
Chest wall retractions	Skin sucking in between ribs, at neck, or below ribcage	Moderate–severe distress
Cyanosis	Blue discolouration of lips, tongue, fingernails	Severe hypoxia — emergency
Unable to speak	Cannot complete a full sentence due to breathlessness	Severe — near-emergency
Altered consciousness	Confusion, agitation, drowsiness in a patient with breathing difficulty	Pre-arrest — emergency
Silent chest	No breath sounds on auscultation or no visible chest movement	Life-threatening — emergency

2. AIRWAY MANAGEMENT — LOW-RESOURCE

STEP 1	OPEN THE AIRWAY
	Tilt head back and lift chin (head-tilt chin-lift). In trauma: jaw thrust without head tilt to protect spine.
STEP 2	CLEAR THE AIRWAY
	Remove visible foreign material with finger sweep. Roll patient onto side to drain blood or vomit.
STEP 3	RECOVERY POSITION
	Unconscious patient breathing: roll onto side. This prevents airway obstruction from tongue and aspiration of vomit.
STEP 4	POSITIONING FOR BREATHLESSNESS
	Conscious patient who is breathless: sit upright, leaning slightly forward (tripod position). Never lay flat.
STEP 5	OXYGEN
	Administer high-flow oxygen if available. Fan the face if no oxygen — mild improvement in perceived breathlessness.

3. ASTHMA

3.1 Recognition

Feature	Mild–Moderate Attack	Severe / Life-Threatening Attack
Speech	Can speak in sentences	Single words only or cannot speak
Breathing rate	Increased but < 25/min	> 25/min
Wheeze	Present — expiratory wheeze	Silent chest — no wheeze = very severe
Position	Can lie down / walk	Unable to lie down; hunched forward
Consciousness	Alert	Confused, drowsy, or agitated
Lips/nails	Normal colour	Blue (cyanosis) — emergency

3.2 Management

- Sit the patient upright — forward-leaning position reduces work of breathing.
- Remove from trigger: smoke, dust, animals, cold air, chemical fumes.
- Calm the patient — anxiety worsens bronchospasm.
- Salbutamol inhaler (if available): 4–8 puffs via spacer or improvised spacer (plastic bottle). See MSF protocols for dosing.
- Improvised spacer: cut the bottom off a 500 mL plastic bottle; insert inhaler in mouthpiece end; breathe through bottle neck.
- Life-threatening attack (silent chest, cyanosis, confusion): emergency referral — requires IV bronchodilators.
- Triggers to avoid long-term: smoke, cold air, dust, strong perfumes, exercise in cold air, respiratory infections.

4. PNEUMONIA

4.1 Recognition

Sign/Symptom	Description	Significance
Cough	Productive cough with yellow, green, or rust-coloured sputum	Common presentation
Fever	Temperature > 38°C	Present in most bacterial pneumonia
Rapid breathing	Adult: > 20/min. Child: > 50/min (< 2 months); > 40/min (2–12 months); > 30/min (1–5 years)	Most reliable clinical sign
Chest pain	Sharp pain worse on breathing or coughing (pleuritic)	Indicates pleural involvement
Reduced air entry	Breath sounds reduced or absent on one side — dullness on percussion	Consolidation or effusion
Confusion (elderly)	New onset confusion in elderly patient with fever and cough = pneumonia until proven otherwise	High mortality risk

4.2 Severity Assessment — CURB-65 (Simplified)

- C — Confusion: new onset confusion
- U — Urea > 7 mmol/L (lab not always available)
- R — Respiratory rate ≥ 30 /min
- B — BP systolic < 90 or diastolic ≤ 60 mmHg
- 65 — Age ≥ 65 years
- Score 0–1: likely manageable with outpatient antibiotics. Score ≥ 2 : consider inpatient care. Score ≥ 3 : severe — urgent care needed.

4.3 Low-Resource Supportive Management

- Position upright or semi-recumbent — improves ventilation and reduces aspiration risk.
- Hydration: encourage oral fluids — prevents dehydration and helps liquefy secretions.
- Breathing exercises: deep breathing and coughing exercises every 2 hours helps clear secretions.
- Fever management: cooling with damp cloths; oral paracetamol if available.
- Antibiotics are required — prescribe per MSF/WHO guidelines for pneumonia. Supportive care alone is insufficient for bacterial pneumonia.
- Monitor breathing rate every 4–6 hours — worsening rate indicates treatment failure.
- Refer any patient who cannot take oral fluids, has cyanosis, or CURB-65 ≥ 2 .

5. TENSION PNEUMOTHORAX — LIFE-THREATENING EMERGENCY

5.1 Recognition

A tension pneumothorax occurs when air accumulates under pressure in the pleural space, collapsing the lung and compressing the heart and great vessels. Death follows within minutes if untreated.

Sign	Description
Absent breath sounds	No breath sounds on one side of the chest
Tracheal deviation	Trachea deviated AWAY from the affected side (late sign — may not be present)
Hypotension + distended neck veins	Shock with raised jugular venous pressure
Increasing respiratory distress	Rapidly worsening breathlessness, cyanosis, altered consciousness
Mechanism	Penetrating chest wound, blunt trauma, rib fractures, blast injury

5.2 Emergency Management — Needle Decompression

Needle decompression is life-saving and can be performed in the field with basic equipment when tension pneumothorax is suspected clinically.

STEP 1	CONFIRM CLINICALLY
	Do NOT wait for imaging. Clinical signs of tension pneumothorax = act immediately.
STEP 2	POSITION
	Patient semi-recumbent or sitting if possible.
STEP 3	NEEDLE DECOMPRESSION
	Insert large-bore needle (14–16 G if available) into 2nd intercostal space, mid-clavicular line, superior border of the rib (avoid neurovascular bundle below each rib). Rush of air = confirms diagnosis.
STEP 4	SEAL AND REFER
	Apply occlusive dressing. Arrange immediate surgical referral for definitive chest drain.

6. OPEN CHEST WOUNDS (SUCKING CHEST WOUND)

An open chest wound allows air to enter the pleural space with each breath, collapsing the lung on that side. It can rapidly progress to tension pneumothorax.

STEP 1	SEAL THREE SIDES
	Apply an occlusive dressing (plastic wrap, foil, clean plastic bag) and tape THREE sides only, leaving one side open. This creates a one-way valve — air exits on expiration but cannot enter on inspiration.

STEP 2	MONITOR BREATHING
	Watch for signs of tension pneumothorax developing (worsening distress, tracheal deviation). If these develop, briefly lift the unsealed edge to release pressure.
STEP 3	POSITION
	Sit patient upright. Keep affected side down if possible.
STEP 4	REFER URGENTLY
	All open chest wounds require surgical management.

7. RIB FRACTURES & FLAIL CHEST

7.1 Single / Multiple Rib Fractures

- Pain localised to fracture site, worsened by breathing, coughing, or movement.
- Patient may splint the chest (hold side and breathe shallowly) — this increases pneumonia risk.
- Management: analgesia to allow deep breathing; encourage deep breaths and coughing every hour.
- Circumferential chest binding is CONTRAINDICATED — restricts breathing and worsens outcome.
- Refer if > 3 ribs fractured, elderly patient, or any respiratory compromise.

7.2 Flail Chest

Flail chest occurs when multiple adjacent ribs are broken in two places, creating a free-floating segment. This segment moves paradoxically (inward on inspiration, outward on expiration), severely impairing ventilation.

- Recognition: paradoxical chest wall movement visible on examination.
- Apply gentle manual stabilisation over the flail segment if resources allow.
- Position: injured side down if possible — splints the flail segment naturally.
- High-flow oxygen; analgesia; urgent referral — requires positive pressure ventilation in severe cases.

8. RED FLAGS — CRITERIA FOR URGENT REFERRAL

Red Flag Criterion	Clinical Significance / Action
Silent chest / no breath sounds	Severe asthma or tension pneumothorax — life-threatening emergency
Cyanosis	Severe hypoxia — oxygen immediately; urgent referral
Respiratory rate < 10 or > 30	Impending respiratory failure
Tension pneumothorax signs	Absent breath sounds + shock + distended neck veins — immediate needle decompression
Open chest wound	Immediate three-sided seal; urgent surgical referral
Flail chest	Paradoxical movement — urgent ventilatory support
Altered consciousness + breathing	Pre-arrest — airway management and emergency referral
Aspiration of blood or vomit	Suction if available; recovery position; urgent referral

REF. REFERENCES

- [1] MSF Clinical Guidelines — Respiratory Emergencies. MSF, 2022. <https://medicalguidelines.msf.org>
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- [6] WHO. Acute Respiratory Infections in Children: Case Management in Small Hospitals. Geneva: WHO.

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